WRDS Database Structure, Linking & Portfolio Construction

CRSP, Compustat, IBES, RavenPack, and Fama–French Factor Replication

FIN 4080, Behavioral Finance

WRDS Database Structure, Linking & Portfolio Construction

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Market Data

CRSP

PERMNO, returns,
delisting adjustments

Fundamentals

Compustat

GVKEY, book equity,
financial statements

Expectations

I/B/E/S

Analyst forecasts,
GAAP vs. Non-GAAP, quality

Alternative Data

RavenPack

News sentiment,
event taxonomy

Integration

Linking

CCM table, temporal
alignment, primary shares

Application

Fama–French

2×3 sorts,
SMB/HML replication

CRSP: What is Inside?

Coverage

- Time: from 1925 to present.
- Universe: NYSE (1925), AMEX (1962), NASDAQ (1972), NYSE Arca (2006), Cboe BZX (2012).
- Frequency: daily and monthly.
- Inclusion: securities from IPO; delisting returns recorded.

Key Data Items (Monthly)

- `PRC`: closing price (negative if bid/ask average).
- `RET`, `RETX`: total return and return excluding dividends.
- `SHROUT`: shares outstanding (thousands).
- `DLRET`: delisting return.
- `EXCHCD`, `SHRCD`: exchange and share type codes.

CRSP: Identifier System

- **PERMNO**: permanent *security* identifier. One firm can have multiple PERMNOs (e.g., multiple share classes).
- **PERMCO**: permanent *company* identifier. Aggregates all securities under one corporate entity.
- **Ticker**: Not stable – tickers are reused, change over time, and differ across exchanges.
- **CUSIP**: changes with corporate actions (M&A, spin–offs, share class changes); Compustat historically kept only the most recent CUSIP, making historical matching difficult.

Rule of thumb: never merge on ticker or CUSIP for historical panels; always use PERMNO / PERMCO.

Example: Alphabet (Google) in CRSP

	CUSIP Header	PERMNO	PERMCO	Nasdaq Issue Number	Exchange Code Header	Standard Industrial Classification Code Header	Date of Observation	Bid or Low Price	Ask or High Price	Price or Bid/Ask Average	Volume	Returns	Bid	Ask	Shares Outstanding
322	02079K30	90319	45483	44290	3	7375	20220930	95.65000	110.86000	95.65000	6122816	-0.116152	95.67000	95.68000	5978000
323	02079K10	14542	45483	74070	3	7375	20221031	92.60000	104.93000	94.66000	5836868	-0.015497	94.65000	94.67000	6086000
324	02079K30	90319	45483	44290	3	7375	20221031	92.22000	104.48000	94.51000	6806717	-0.011918	94.52000	94.53000	5973000
325	02079K10	14542	45483	74070	3	7375	20221130	83.49000	101.45000	101.45000	6026212	0.071730	101.44000	101.45000	6086000
326	02079K30	90319	45483	44290	3	7375	20221130	83.43000	100.99000	100.99000	7157998	0.068564	101.01000	101.02000	5973000
327	02079K10	14542	45483	74070	3	7375	20221230	86.46000	101.28000	88.73000	4984841	-0.125382	88.73000	88.76000	5968000
328	02079K30	90319	45483	44290	3	7375	20221230	86.02000	100.99000	88.23000	6018502	-0.126349	88.22000	88.24000	5956000
329	02079K10	14542	45483	74070	3	7375	20230131	86.77000	101.21000	99.87000	5253124	0.125549	99.88000	99.91000	5968000

- Multiple PERMNOs under one PERMCO

CRSP Adjustment Factors: CFACPR and CFACSH

Why Adjustment Factors?

- Corporate actions (stock splits, reverse splits, stock dividends, spin-offs) change the *nominal* price and shares outstanding without changing the firm's total market value.
- Without adjustment, a 2-for-1 split would appear as a 50% price drop—not a real loss.

CRSP Fields

- `CFACPR`: Cumulative Factor to Adjust Prices.
- `CFACSH`: Cumulative Factor to Adjust Shares Outstanding.
- Both are always one for latest CRSP observations (backfill historical data).

Key Insight

$$\textit{Adjusted Price} = \frac{PRC}{CFACPR}, \quad \textit{Adjusted Shares} = SHROUT \times CFACSHR \tag{1}$$

Example: NVDA 4-for-1 split

	PERMNO	Date of Observation	Price or Bid/Ask Average	Volume	Returns	Shares Outstanding	Cumulative Factor to Adjust Prices	Cumulative Factor to Adjust Shares/Vol
267	86580	20210226	548.58002	1666380	0.055794	620000	4	4
268	86580	20210331	533.92999	1906338	-0.026414	622345	4	4
269	86580	20210430	600.38000	1675421	0.124455	623000	4	4
270	86580	20210528	649.78003	1760518	0.082281	623000	4	4
271	86580	20210630	800.09998	2562882	0.231586	623000	4	4
272	86580	20210730	194.99001	3711893	-0.025172	2500000	1	1
273	86580	20210831	223.85001	6403341	0.148213	2500000	1	1
274	86580	20210930	207.16000	4992551	-0.074559	2500000	1	1
275	86580	20211029	255.67000	4963403	0.234167	2502000	1	1
276	86580	20211130	326.76001	10785582	0.278054	2500000	1	1
277	86580	20211231	294.10999	11041329	-0.099798	2500000	1	1

Note the sudden drop in nominal price and the corresponding adjustment in `CFACPR` to ensure return continuity. **Latest** CFACPR (CFACSHR) is always **1** (backfill old data if splits).

Delisting Returns: Why and How

What is Delisting?

- A security is removed from the exchange (merger, liquidation, failure to meet listing standards, bankruptcy).
- The last trading price may not reflect the final cash or stock distribution received by shareholders.
- `DLRET` in `crsp.msedelist` captures the return from the last trading price to the final settlement value.

The Delisting Bias Problem

- Shumway (1997) and Shumway & Warther (1999) showed that ignoring delisting returns creates a severe downward bias, especially for small firms with high delisting rates.
- CRSP's monthly `RET` ends at the last trading month; without `DLRET`, the investor's total return is understated.

Computing the Adjusted Return

Basic Formula

$$R_t^{\text{adj}} = (1 + R_t) \times (1 + \text{DLRET}_t) - 1$$

CRSP `DLRET` Codes

- **Numeric value**: standard return (e.g., -0.30, $+0.15$).
- **Missing (`.`)**: no delisting return available; treat as 0.

Missing Value Handling (SAS Convention)

- `sum(1, ret)` returns 1 if `ret` is missing; `1 + ret` returns missing.
- Standard construction:

$$\texttt{retadj} = \texttt{sum(1, ret)} \times \texttt{sum(1, dlret)} - 1$$

This safely treats missing `DLRET` as 0 without losing observations.

From Market Prices to Fundamental Data

"Prices reflect expectations, but expectations about what?"

- CRSP tells us **what** the market is pricing and **how** it has performed.
- To understand **why** prices move, we need the underlying economics of the firm: assets, earnings, leverage, and growth.
- This requires linking market data to accounting data.



Compustat provides the fundamental balance-sheet and income-statement data necessary to anchor market valuations in firm fundamentals.

Compustat: What is Inside?

Coverage

- Time: from 1950 to present.
- Universe: U.S. and Canadian firms filing 10-K, 10-Q includes NYSE, NASDAQ, AMEX, regional exchanges, OTC, and certain bond–only issuers.
- Frequency: quarterly and annual.
- Inclusion: traditionally, firms appear only after several years of public disclosure (survivorship bias concern in early years).

Key Data Items (Annual)

- `AT`: total assets; `SEQ`: stockholders' equity.
- `PSTK`, `PSTKL`, `PSTKRV`: preferred stock (par, liquidation, redemption).
- `TXDITC`: deferred taxes and investment tax credit.
- `NI`: net income.

Compustat: Identifier System

- **GVKEY**: permanent *company* identifier (Global Company Key).
- **IID**: issue identifier; combined with GVKEY gives a *security–level* ID (`GVKEY` + `IID`).
- **DATADATE**: reporting date (usually fiscal year–end).
- **FYEAR**: fiscal year label (calendar year minus 1 if fiscal year ends Jan–May).
- Standard filters:
  - `INDFMT='INDL'`: industrial format (vs. financial services).
  - `DATAFMT='STD'`: standardized presentation.
  - `POPSRC='D'`: domestic (U.S., Canada, ADRs).
  - `CONSOL='C'`: consolidated statements.

Example: Alphabet (Google) in Compustat

Global Company Key	Data Date	Data Year – Fiscal	Assets – Total	Net Income (Loss)	Preferred Stock – Redemption Value	Stockholders Equity – Parent	Deferred Taxes and Investment Tax Credit	Header Primary Global Issue ID
160329	20031231	2003	871.4580	105.6480	40.8150	602.6410	18.5100	01
160329	20041231	2004	3313.3510	399.1190	0.0000	2929.0560	0.0000	01
160329	20051231	2005	10271.8130	1465.3970	0.0000	9418.9570	35.4190	01
160329	20061231	2006	18473.3510	3077.4460	0.0000	17039.8400	40.4210	01
160329	20071231	2007	25335.8060	4203.7200	0.0000	22689.6790	0.0000	01
160329	20081231	2008	31767.5750	4226.8580	0.0000	28238.8620	12.5150	01

Compustat aggregates all share classes under a single GVKEY, but security-level detail requires `GVKEY + IID`.

Beyond Historical Fundamentals

- Compustat provides a **backward-looking** snapshot: what the firm *has already done*.
- Asset prices, however, are forward-looking. They reflect investor expectations about *future* earnings, cash flows, and growth.
- Where do these expectations come from? One major source is the consensus of professional equity analysts.

$$\Downarrow$$

I/B/E/S captures the collective forward-looking intelligence of the sell-side analyst community.

What is I/B/E/S?

Institutional Brokers' Estimate System (I/B/E/S)

- The conventional source for analyst earnings forecasts, recommendations, and target prices in academic finance.
- Contributed by 3,000+ broker houses worldwide, covering 30,000+ individual analysts.
- U.S. coverage begins in 1975; international coverage from 1987.
- Data collected daily as analysts release forecasts; summary consensus computed monthly.

Three Main Data Products

1. **Estimates**: Historical analyst forecasts (EPS, revenue, EBITDA, price targets, etc.). Available at detailed (analyst–level) and consensus (summary) levels.
2. **Recommendations**: Buy / hold / sell ratings.
3. **Guidance**: Management's own forecasts (formerly Company Issued Guidance + First Call).

Example: Analyst Report for Amazon

CIBC World Markets

Equity Research
Company Update

April 5, 2005

Internet & E-Commerce

Stock Rating:
Sector Performer

Sector Weighting:
Overweight

12-18 mo. Price Target	None
AMZN-OTC (4/5/05)	$35.28
Key Indices: NASDAQ, GSTI Comp, NAS100	
3-5-Yr. EPS Gr. Rate (E)	20.0%
52-week Range	$32.82-$54.70
Shares Outstanding	425.0M
Float	298.0M Shrs
Avg. Daily Trading Vol.	8,315,000
Market Capitalization	$14,994.0M
Dividend/Div Yield	Nil / Nil
Fiscal Year Ends	December
Book Value	NM
2005 ROE (E)	NM
LT Debt	$1.9B
Preferred	Nil
Common Equity	Nil
Convertible Available	Yes

Earnings per Share	Prev	Current
2004		$0.93A
2005		$1.26E
2006		$1.54E
P/E		
2004		37.9x
2005		28.0x
2006		22.9x

Note: EPS Estimates are Untaxed

Amazon.com

AMZN Buys BookSurge & Takes Another Step in Changing Book E-Tailing

- Earlier this week, AMZN purchased privately-held BookSurge, an inventory-free, "print-on-demand" book fulfillment company. Terms of the deal were not disclosed. However, our channel checks suggest BookSurge had sales of ~$4M in '04.
- Although the transaction has an immaterial impact on AMZN's financials, BookSurge strengthens AMZN's ability to meet book demand along the farther end of the "long tail", as well as strengthens its fulfillment capabilities in international markets (e.g., foreign language translations).
- BookSurge's "print-on-demand" technology enables Amazon to meet demand for dated, niche, and out-of-print titles, while also integrating AMZN's long-term goals for "search inside the book". BookSurge also had leadership in publishing books in various foreign languages.
- Assuming a reasonable price, the deal is consistent with AMZN's strategy. More importantly, in our view, it suggests AMZN continues to raise the bar for its competitors and provides yet another datapoint in developing a long-term view of how e-commerce and retailing evolves, one vertical at a time.

Stock Price Performance

Amazon.com (AMZN)

Example: Analyst Report for Amazon

- A typical detailed analyst report includes:
  - EPS estimates for current and future fiscal years
  - Revenue and EBITDA projections
  - Buy/hold/sell recommendation with target price
  - Qualitative analysis of business segments
- These individual forecasts are aggregated by I/B/E/S into consensus statistics.
- The **detail file** preserves the analyst identity and broker; the **summary file** provides the consensus mean, median, and dispersion.

I/B/E/S Data Dimensions

Dimension	Description
Detailed vs. Consensus	Individual analyst forecasts vs. monthly summary statistics
EPS vs. Non–EPS	Earnings per share vs. revenue, EBITDA, price targets, etc.
Adjusted vs. Non–Adjusted	Split–adjusted (rounded) vs. as–reported (unadjusted)
U.S. vs. Non–U.S.	Domestic and international coverage

Key Point: the "easiest–to–use" adjusted consensus files are **rounded to two decimals** and subject to stock–split adjustment timing differences. This matters for earnings surprise calculations.

Identifier System

I/B/E/S Internal Identifiers

- **TICKER**: I/B/E/S's own permanent security identifier. Stable throughout I/B/E/S history; **not** the exchange ticker.
- **CUSIP / SEDOL**: Historical CUSIP (or SEDOL when CUSIP unavailable). Can change over time; not suitable as a permanent merge key.
- **ESTIMATOR**: Broker house code (e.g., Goldman Sachs, Merrill Lynch).
- **ANALYS**: Individual analyst code within a broker.

Linking to CRSP and Compustat

- Use the I/B/E/S–CRSP linking table (available on WRDS) or CUSIP–based matching.
- CUSIP changes with splits and corporate actions; for historical panels, prefer the official linking reference.
- Note: I/B/E/S ticker is not CRSP ticker; always verify with CUSIP or the dedicated link table.

Key Variables and Jargon

Forecast Content

- **VALUE**: the analyst's point estimate (e.g., EPS forecast of \$5.80).
- **ACTUAL**: realized earnings announced by the firm.
- **MEASURE**: type of forecast (EPS, revenue, EBITDA, etc.).

Share Base Indicators

- **PDI** (Primary / Diluted Indicator): share base selected for the company.
- **PDF** (Primary / Diluted Flag): share base selected for the specific estimate.

Forecast Period Indicator (FPI)

- Codes identifying the horizon of the estimate.
- Example: `FPI = 1` = next fiscal year; `FPI = 6` = next fiscal quarter.
- Critical for aligning forecasts with the correct earnings announcement.

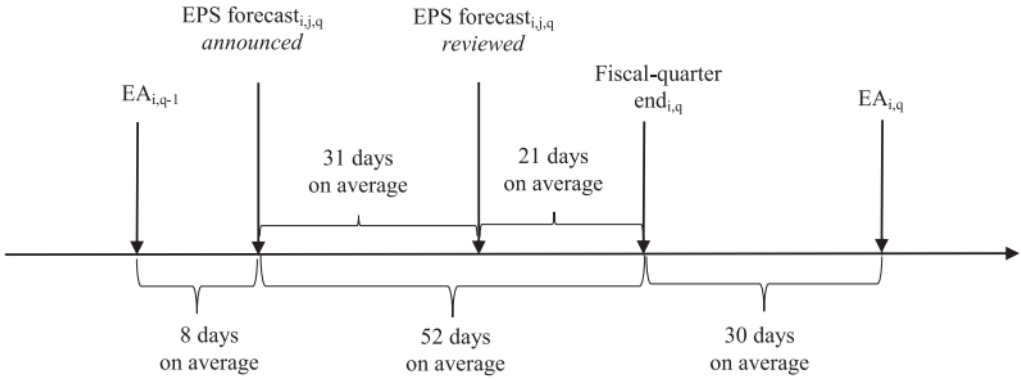
Forecasting Timeline: Key Dates

Variable	Definition
ANNDATS	Date the forecast or actual was *announced* (publicly released)
ACTDATS	Date the forecast or actual was *recorded* by I/B/E/S
FPEDATS	Forecast Period End Date: the fiscal date to which the estimate applies
REVDATS	Most recent date the estimate was confirmed as accurate
STATPERS	Statistical Period: the calendar date on which consensus stats are calculated
ANNDATS_ACT	Date the actual earnings were announced

Important Distinction

- `ANNDATS` $\neq$ `ACTDATS`: a forecast may be announced on Jan 17 but entered into the database on Jan 18.
- For event studies, use `ANNDATS` (the market's information set); for database integrity checks, use `ACTDATS`.
- `STATPERS` is a month–end date in the summary file; all consensus statistics (mean, median, std dev, number of estimates) are computed as of this date.

Forecasting Timeline: Example



- Reference: Kaplan, et al (2019): Truncating optimism

Example: Analyst Estimates for Alphabet, Detail file, Q1, 2026

	Official Ticker Symbol	IBES Ticker Symbol	Measure (Data Type Indicator)	Announce Date, SAS Format	Forecast Period Indicator	CUSIP/SEDOL	Activation Date, SAS Format	Review Date, SAS Format	Contributing estimator code	Analyst Code	Estimate Value	PF(L	Actual Value, from the Detail Actuals File	Activation date of the Actual, from the Actuals File
1006	GOOG	GOOG/1	EPS	2026-02-05	6	02079K10	2026-02-05	2026-04-24	00176	120988	2.64	D	2.62	2026-04-29
1007	GOOG	GOOG/1	EPS	2026-02-05	6	02079K10	2026-02-05	2026-04-27	04004	000000	2.34	D	2.62	2026-04-29
1008	GOOG	GOOG/1	EPS	2026-02-04	6	02079K10	2026-02-06	2026-02-06	03477	119704	2.55	D	2.62	2026-04-29
1009	GOOG	GOOG/1	EPS	2026-02-05	6	02079K10	2026-02-06	2026-02-06	04235	135458	2.55	D	2.62	2026-04-29
1010	GOOG	GOOG/1	EPS	2026-02-05	6	02079K10	2026-02-06	2026-02-08	00873	142116	2.69	D	2.62	2026-04-29
1011	GOOG	GOOG/1	EPS	2026-02-05	6	02079K10	2026-02-06	2026-04-24	00898	078750	2.66	D	2.62	2026-04-29
1012	GOOG	GOOG/1	EPS	2026-02-05	6	02079K10	2026-02-06	2026-04-27	01267	000000	2.67	D	2.62	2026-04-29
1013	GOOG	GOOG/1	EPS	2026-02-04	6	02079K10	2026-02-08	2026-04-21	00238	022953	2.74	D	2.62	2026-04-29
1014	GOOG	GOOG/1	EPS	2026-02-04	6	02079K10	2026-02-08	2026-04-24	00098	080183	2.6	D	2.62	2026-04-29

- FDI = 6 implies next fiscal quarter end

Example: Analyst Estimates for Alphabet, summary file, Q1, 2026

	IBES Ticker Symbol	CUSIP/SE	Offici Ticker Symbol	Company Name	I/B/E/S Statistical Period (SAS Format)	Measu	Per	Fore Peri Indi	Estim Flag (P=Pr S=Sec	Numb of Esti	Median Estimate	Mean Estimate	Standard Deviation	High Estimate	Low Estimate	Forecast Period End Date (S
248	GOOG/1	02079K10	GOOG	ALPHABET INC	15JAN2026	EPS	QTR	6	P	32	2.61	2.63	0.12	3	2.5	31DEC2025
249	GOOG/1	02079K10	GOOG	ALPHABET INC	19FEB2026	EPS	QTR	6	P	31	2.6	2.6	0.11	2.8	2.3	31MAR2026
250	GOOG/1	02079K10	GOOG	ALPHABET INC	19MAR2026	EPS	QTR	6	P	32	2.6	2.6	0.11	2.8	2.3	31MAR2026
251	GOOG/1	02079K10	GOOG	ALPHABET INC	16APR2026	EPS	QTR	6	P	32	2.61	2.69	0.47	5.2	2.3	31MAR2026
252	GOOG/1	02079K10	GOOG	ALPHABET INC	14MAY2026	EPS	QTR	6	P	30	2.9	2.88	0.12	3.1	2.6	30JUN2026

GAAP vs. Non–GAAP Earnings

Two Reporting Standards

- **GAAP**: standardized under U.S. accounting rules; includes all operational and non–operational items.
- **Non–GAAP (Pro Forma)**: excludes "non–recurring" or "non–cash" items (restructuring charges, stock–based compensation, write–downs, etc.).

Why It Matters

- Wall Street analysts typically forecast **non–GAAP** earnings.
- In 2017, 97% of S&P 500 firms reported non–GAAP metrics (up from 59% in 1996).
- The same firm can have very different GAAP and non–GAAP EPS numbers; comparing analyst forecasts to GAAP actuals creates measurement error.

Regulation G (2003)

- SEC requires: (1) label as "non–GAAP," (2) report GAAP first, (3) provide detailed reconciliation.
- Did not eliminate pro forma reporting; analysts still forecast non–GAAP.

GAAP vs. Non-GAAP Earnings: TSLA 10-Q, Q2, 2026

In millions of USD or shares as applicable, except per share data	Q2-2025	Q3-2025	Q4-2025	Q1-2026	Q2-2026
Net income attributable to common stockholders (GAAP)	**1,172**	**1,373**	**840**	**477**	**1,114**
Stock-based compensation expense, net of tax	443	459	682	803	989
Digital assets unrealized (gain) loss, net of tax	(222)	(62)	239	173	87
SpaceX equity investment unrealized gain, net of tax	-	-	-	-	(763)
Certain tax items [1]	-	-	-	-	(274)
Net income attributable to common stockholders (non-GAAP)	**1,393**	**1,770**	**1,761**	**1,453**	**1,153**
Less: Buy-out of noncontrolling interest	-	-	-	-	(2)
Net income used in computing diluted EPS attributable to common stockholders (non-GAAP)	**1,393**	**1,770**	**1,761**	**1,453**	**1,155**

The Adjusted vs. Unadjusted Problem

I/B/E/S "Adjusted" Data

- I/B/E/S applies its own cumulative split–adjustment factors (`CFACSHR`) to normalize forecasts across stock splits.
- Adjusted consensus files are **rounded to two decimal places**.

The Rounding Bias (Payne & Thomas, 2003)

- Rounding is most pronounced for: larger firms, high M/B firms, better performers.
- Can create artificial clustering at zero forecast errors and distort earnings management inferences.
- Example: a split–adjusted mean of \$1.01 vs. actual \$0.99 may round to identical values, masking a real surprise.

The Split–Date Mismatch

- I/B/E/S effective split date is the date the split became "effective" *inside the IBES database*—not necessarily the true CRSP split date.
- Using CRSP's `CFACSH` and the true split date is more accurate for back–adjusting unadjusted data.

Data Quality Issue 1: Rewriting History

Ljungqvist, Malloy & Marston (JF, 2009)

- Compared seven I/B/E/S downloads (2000–2007).
- Found 1.6%–21.7% of matched recommendation observations differed between versions.
- Four types of changes: alterations, deletions, additions, anonymizations.
- Implications: trading signal profitability, analyst performance persistence, and consensus change measures are sensitive to the vintage used.

Call et al. (2020): Vanishing Estimates

- **Detailed file**: 11.68% of 2009 estimates missing in 2015; 6.01% of 2015 estimates missing in 2009.
- **Summary file**: far more stable (0.11% / 1.49% missing).
- Broker contractual rights allow retroactive removal of their analysts' forecasts.
- Some brokers (e.g., Goldman Sachs) provide estimates only to summary files, never to detailed files.

Data Quality Issue 2: Encrypted History

MiFID II and Anonymization (October 2018)

- Regulatory compliance required I/B/E/S to anonymize broker and analyst identifiers for many contributors.
- 88 contributor names removed from detailed estimates data; UBS estimates fully removed.
- I/B/E/S informed WRDS that broker and analyst IDs will continue to be **reshuffled without notice**.

Pierson (WRDS, 2020)

- Compared 2014 vs. 2019 detailed files: 13.8% of `ESTIMATOR` codes modified; up to 30.7% of `ANALYS` codes changed.
- Many analyst ID changes are *unrelated* to the affected brokers, suggesting systematic re–coding.

Scope of Impact

- Affects **detailed estimates only** (due to regulatory compliance).
- **Summary / consensus files remain unchanged**

Best Practices for Using I/B/E/S

1. **Prefer unadjusted consensus data** over adjusted detailed data for earnings surprise calculations.
2. **Apply CRSP split factors** (`CFACSH`) yourself rather than relying on I/B/E/S's internal adjustment timing, which may not match true split dates.
3. **Use the summary file** when possible: it is more stable across vintages and immune to analyst–ID reshuffling.
4. **Be explicit about GAAP vs. non–GAAP**: analyst forecasts are typically non–GAAP; match them to non–GAAP actuals or use the reconciliation provided in 8–K exhibits.
5. **Document your vintage**: if using detailed files, state the download date. Results may not replicate across different I/B/E/S vintages.
6. **For event studies**: use `ANNDATS` (public announcement date), not `ACTDATS` (database entry date), to align with market reactions.
7. **For linking**: use the official I/B/E/S–CRSP link table or CUSIP with date bounds; never assume I/B/E/S ticker equals CRSP ticker.

The Information Spectrum

- I/B/E/S provides **structured, periodic** expectations: quarterly EPS forecasts, annual target prices, discrete recommendations.
- But firm value is affected daily by a continuous flow of **unstructured** information: news stories, product recalls, ESG controversies, macro shocks, and social-media sentiment.
- Traditional databases (CRSP, Compustat, I/B/E/S) do not capture the *timeliness* and *nuance* of text-based information.



RavenPack bridges this gap by converting unstructured news text into structured, machine-readable sentiment and event analytics in real time.

What is RavenPack?

RavenPack News Analytics

- Leading provider of real–time news analytics and sentiment data for financial research and quantitative investing.
- Available on WRDS since 2000 (U.S. firm–level coverage); 20+ years of millisecond–precision historical archives.
- Processes unstructured text from 40,000+ curated sources in 13 languages: Dow Jones, WSJ, Benzinga, MT Newswires, regulatory wires, and thousands of web publications.
- 100% machine–readable output: each news item is structured into entity–level records with sentiment, relevance, and event classifications.

Academic Use Cases

- Market efficiency, behavioral finance, volatility forecasting, ESG controversy detection, and event–driven strategies.
- Complements CRSP prices and Compustat fundamentals by adding a **real–time information layer**.

RavenPack Data Products on WRDS

1. **RPA 1.0 – Equities**
   - Company–level analytics: sentiment, relevance, novelty, and event detection for 90,000+ public and private firms.
   - Each record contains 46 fields including timestamp, entity identifiers, sentiment scores, and event taxonomy.
2. **RPA 1.0 – Global Macro**
   - Macro–level analytics: currencies, commodities, places, persons, organizations (160,000+ macro entities).
   - Each record contains 34 fields; useful for studying geopolitical risk, monetary policy, and commodity shocks.
3. **RPA 1.0 – Mapping Files**
   - Entity mapping dataset: links RavenPack entity IDs to external identifiers (CRSP PERMNO, Compustat GVKEY, CUSIP, ISIN).
   - Event taxonomy dataset: hierarchical classification of 7,000+ detected event types.
   - Source dataset: metadata on all 40,000+ news providers (coverage dates, trustworthiness, influence).

Core Analytics: Sentiment, Relevance, and Events

Event Sentiment Score (ESS)

- Scale: 0 (most negative) to 100 (most positive); 50 is neutral.
- Derived from multiple NLP algorithms applied to the full text of each news item.
- Assigned at the **entity–level**: if one article mentions Apple and Samsung, each gets its own ESS.

Relevance Score

- Scale: 0 to 100; measures how strongly the entity is related to the underlying news story.
- Average relevance for firm–level news is above 94, indicating high precision in entity detection.

Novelty Score

- Measures whether the news item contains new information or is a repetition of previously reported facts.
- Critical for distinguishing genuine information shocks from stale re–broadcasts.

Event Taxonomy and Data Structure

Hierarchical Event Classification (6 Levels)

Level	Example
Topic	Corporate & Market
Group	Earnings & Dividends
Type	Earnings Announcement
Sub–type	Earnings Beat
Property	Revenue
Category	Product Segment

Entity–Level Records

- One news story $\Rightarrow$ multiple entity–level records (one per mentioned entity).
- Entity types: Company, Organization, Currency, Commodity, Place, Nationality, People, Product, Sports Team.
- Each record carries: `TIMESTAMP_UTC`, `ENTITY_ID`, `ENTITY_NAME`, `ENTITY_TYPE`, `ESS`, `RELEVANCE`, `NOVELTY`, `EVENT_TYPE`, and unique story identifiers.

Linking RavenPack to CRSP and Compustat

Entity Mapping

- RavenPack assigns a permanent `ENTITY_ID` to each tracked entity.
- The RPA 1.0 Mapping Files provide cross–walks to:
  - CRSP: PERMNO, PERMCO (via CUSIP or company name matching).
  - Compustat: GVKEY, IID (via CUSIP/ISIN or entity name resolution).
  - Other identifiers: ISIN, SEDOL, LEI.

Linking Challenges

- **Entity disambiguation**: "Apple" could be Apple Inc. (company) or the fruit (commodity). RavenPack's NLP resolves this using context and industry classification.
- **Temporal alignment**: news timestamps are in UTC milliseconds; align with CRSP trading dates using the nearest calendar date.
- **Multiple mentions**: a firm may appear multiple times in one article (e.g., acquirer and target in an M&A story). Each mention gets a separate record with its own relevance and sentiment.

Best Practice: filter on `RELEVANCE >= 90` and `ENTITY_TYPE = 'COMPANY'` for firm–level financial research.

Academic Applications and Best Practices

Common Research Designs

- **Event studies**: measure abnormal returns around news events (e.g., earnings surprises, ESG controversies) using RavenPack's event classification and exact timestamps.
- **Sentiment factors**: construct daily or monthly firm–level sentiment indices (average ESS, news volume) and test their predictive power for returns or volatility.
- **ESG controversy monitoring**: track negative ESG events in real time; RavenPack's ESG Controversy Framework scores firms on environmental, social, and governance incidents.
- **Macro–firm spillovers**: link global macro news (monetary policy, geopolitical events) to individual firm returns via the Global Macro dataset.

Data Quality Considerations

- **Source heterogeneity**: premium wires (Dow Jones) vs. blogs and social media have different credibility and speed. Use the Source dataset to weight or filter by trustworthiness.
- **Duplicate stories**: same event reported by multiple outlets. Use the unique story identifier to deduplicate or cluster related articles.
- **Look–ahead bias**: RavenPack timestamps are real–time, but WRDS historical archives are point–in–time. Verify that your extract does not include retroactively corrected scores.

The Integration Challenge

- We have now surveyed four distinct data sources:
  - **CRSP**: security prices and returns
  - **Compustat**: firm fundamentals
  - **I/B/E/S**: analyst forecasts
  - **RavenPack**: real-time news sentiment
- Each uses its own identifier system, its own temporal conventions, and its own entity definitions.
- Before we can use these data together—for example, to construct portfolios conditioned on book-to-market, analyst dispersion, *and* news sentiment—we must solve the linking problem.

$$\Downarrow$$

How do we reliably merge CRSP and Compustat? How do we handle delistings, share classes, and changing identifiers?

Why Linking is Non–Trivial

- **Different universes**: CRSP starts at IPO; Compustat starts when filing history is long enough. Neither is a superset of the other.
- **Different levels**:
  - CRSP is security–centric (PERMNO).
  - Compustat is firm–centric (GVKEY), with security–level detail via IID.
- **Multiple share classes**: one PERMCO (firm) may have multiple PERMNOs; one GVKEY may have multiple IIDs.
- **Identifier instability**: tickers and CUSIPs change; Compustat historically did not retain historical CUSIPs.
- **Time alignment**: fiscal year–end (Compustat) does not align with calendar dates (CRSP).

The Problem with CUSIP

History and Structure

- Introduced in 1964 by the American Bankers Association.
- 6 digits (issuer) + 2 digits (issue) + 1 check digit.

Problems for Longitudinal Research

- **Limited scope**: U.S. and Canada focused (ISIN is global but less common in U.S. databases).
- **Temporal instability**: corporate actions (M&A, spin–offs, reclassifications) alter CUSIPs.
- **Ambiguity**: one firm can have multiple CUSIPs (different share classes, debt issues).
- **Historical gap**: Compustat traditionally stored only the *most recent* CUSIP, making historical matching impossible before the 2025 feed update.

Permanent Identifiers: The Correct Approach

Level	CRSP	Compustat
Security	PERMNO	GVKEY + IID
Company	PERMCO	GVKEY

- **PERMNO**: primary, permanent security ID in CRSP. Tracks a specific traded instrument through its entire life, including exchange changes.
- **PERMCO**: aggregates all PERMNOs under one corporate entity. Use PERMCO when you need firm–level aggregates (e.g., total market capitalization across all share classes).
- **GVKEY**: primary, permanent company ID in Compustat. Stable through name changes, mergers, and restructurings.
- **GVKEY + IID**: required when you need Compustat security–level detail (e.g., specific share class fundamentals).

2025 Compustat Enhancements

New tables in the North America feed (as of 2025)

1. `sec_idhist`
   - Records historical CUSIPs with effective–date ranges.
   - Enables time–varying identifier tracking across corporate actions.
2. `sec_history`
   - Flags a firm's **primary U.S. security** at any point in time.
   - Provides continuity for return–based studies that require the "main equity class."

These tables address the two core historical limitations:

- No historical CUSIP trace $\Rightarrow$ now solved by `sec_idhist`.
- No consistent primary share indicator $\Rightarrow$ now solved by `sec_history`.

CCM: The Official Linking Table

What is CCM?

- **CCM** = CRSP/Compustat Merged *linking reference* (not a merged dataset).
- Table: `crsp.ccmxpf_lnkhist`.
- Provides structured, historical mappings between CRSP and Compustat identifiers.

Key Variables

- `GVKEY`, `LIID`: Compustat identifiers.
- `LPERMNO`, `LPERMCO`: linked CRSP identifiers.
- `LINKDT`, `LINKENDDT`: effective start and end dates of the link.
- `LINKTYPE`: source and reliability of the link.
- `LINKPRIM`: primary issue indicator.

CCM Link Quality Filters

LINKTYPE – Source and Reliability

- `'LU'`: Link research complete. Verified research link. **Highest confidence**.
- `'LC'`: Un–researched link by CUSIP. Less vetted but still widely used.
- Other codes (`'LS'`, `'LX'`, etc.) exist for special cases; standard practice restricts to `'LU'` and `'LC'`.

LINKPRIM – Primary Issue Flag

- `'P'`: Primary issue as identified by Compustat in monthly security data.
- `'C'`: Primary issue assigned by CRSP to resolve overlapping or missing primary markers from Compustat, ensuring one primary security throughout company history.
- `'J'`: Joiner / secondary issue.

Standard filter: `LINKTYPE in ('LU','LC')` and `LINKPRIM in ('P','C')`.

Temporal Alignment in CCM

The Core Condition

```
LINKDT <= DATADATE <= coalesce(LINKENDDT, today())
```

Why this matters

- Links are not perpetual; a GVKEY may map to different PERMNOs over time (M&A, spin–offs, reclassifications).
- Using a static link table without date bounds creates look–ahead bias and mismatches.
- `LINKENDDT` can be missing (`.E` in SAS) for currently active links; coalesce with today's date.

Fiscal vs. Calendar Time

- Compustat `DATADATE` is fiscal year–end.
- CRSP dates are calendar trading dates.
- Portfolio formation in June, year t uses accounting data from fiscal years ending in calendar year $t-1$.

Handling Duplicates and Primary Shares

Duplicate Type 1: One PERMNO, Multiple GVKEYs

- Occurs around M&A events where CRSP and Compustat disagree on entity boundaries.
- Solution: use `LINKPRIM='P'` or `'C'`; inspect `LINKDT` ranges; if necessary, keep the link with the later `LINKDT` or the one corresponding to the surviving entity.

Duplicate Type 2: Multiple Fiscal Year–Ends in One Calendar Year

- Short fiscal years (e.g., change in fiscal year–end) create multiple `DATADATE`s within one calendar year for the same GVKEY.
- Solution: for a given GVKEY and fiscal year, keep the *last* `DATADATE` (`having datadate = max(datadate)`).

CUSIP–Based Merge vs. CCM–Based Merge

Post–1986

- CUSIP–based matching and CCM–based matching yield nearly identical results.
- Differences are negligible for most empirical applications.

Pre–1986 (especially 1970s – early 1980s)

- Noticeable discrepancies due to:
  1. Limited Compustat CUSIP history (Compustat only kept the most recent CUSIP).
  2. Staggered CRSP NASDAQ coverage (NASDAQ added to CRSP in 1972, but backfilling was incomplete).
- For long–term panels spanning pre– and post–1986, CCM remains the gold standard.

Recommendation: always benchmark CUSIP–based results against CCM for early samples.

Merge Sanity Checks

After merging CRSP and Compustat, verify:

1. **No duplicates**: each PERMNO–date or PERMNO–fiscal–year combination should be unique after primary–share filtering.
2. **Market equity sign**: `ME = |PRC| * SHROUT` should be positive; negative `PRC` (bid/ask average) requires absolute value.
3. **Book equity sign**: book equity should be positive for Fama–French construction (negative BE firms are excluded).
4. **Primary share alignment**: for multi–class firms (e.g., Alphabet), ensure that the PERMNO you retain corresponds to the primary share class (the one with the largest or most liquid issue, or the one flagged by `sec_history.PRIHISTUSA`).
5. **Age filter**: early Compustat had a 2–year inclusion lag; excluding firms with age `<= 2` removes survival bias in early years (though this bias is largely gone in modern Compustat daily updates).

Putting Theory into Practice

- We now understand **how** to link CRSP and Compustat reliably using CCM, permanent identifiers, and careful date bounds.
- The next step is to use this merged data to construct classic asset-pricing factors.
- Why? Because replicating the Fama–French factors is the "hello world" of empirical asset pricing: it tests whether you can correctly align accounting data with market data, handle delistings, aggregate multi-class firms, and avoid look-ahead bias.
- Mastering this replication means you can build *any* custom portfolio strategy on WRDS.

$$\Downarrow$$

We now replicate the Fama–French Three-Factor model.

Objective: Replicate Fama–French 1993

Goal

- Construct mimicking portfolios that capture the **Size** (Banz, 1981) and **Value** (Stattman, 1980; Basu, 1977) anomalies.
- Produce monthly time series for:
  - **SMB** (Small Minus Big): return difference between small and large firms.
  - **HML** (High Minus Low): return difference between high and low book–to–market firms.

Key Principle

- All information used to form portfolios must be known *before* the returns they explain.
- This requires careful alignment of accounting disclosure dates and market prices.

Data Requirements: Two Time Points

Market Equity (ME)

- **December of year** $t-1$: used as the denominator for book–to–market ratio.
- **June of year** t: used for size classification and as the portfolio formation date.

Book Equity (BE)

- From Compustat fiscal year ending in calendar year $t-1$.
- Fama–French convention imposes a 6–month gap to ensure the market has access to financial statements by the portfolio formation date.

Returns

- Monthly returns from July of year t to June of year $t+1$ (12–month holding period).
- Delisting returns should be incorporated to avoid delisting bias.

CRSP Universe Filters

Share Codes

- Keep `SHRCD in (10, 11)`: U.S. common shares.
- Exclude ADRs, REITs, ETFs, closed–end funds, preferred stocks, and units.

Exchange Codes

- Keep `EXCHCD in (1, 2, 3)`: NYSE, AMEX, and NASDAQ.
- NYSE breakpoints are used for portfolio splits regardless of this filter; AMEX and NASDAQ stocks are *assigned* to NYSE–determined bins.

Price Requirement

- Firm must have a valid price for December of $t-1$ and June of t.

Compustat: Book Equity (BE) Calculation

Formula (Fama–French 1993)

$$BE = SEQ + TXDITC - PreferredStock$$

Preferred Stock Priority

1. `PSTKRV`: redemption value (if available).
2. `PSTKL`: liquidation value.
3. `PSTK`: par value (book value).
4. If none available, use 0.

Standard Compustat Filters

- `INDFMT='INDL'`, `DATAFMT='STD'`, `POPSRC='D'`, `CONSOL='C'`.

Survivorship Bias

- Early Compustat required firms to have 2+ years of history before inclusion.
- Excluding firms with "age ≤ 2" mitigates this bias in early samples.

CRSP: Market Equity (ME) and Multi–Class Firms

Security–Level ME

$$MEQ = |PRC| \times SHROUT$$

- `PRC` can be negative (bid/ask average when no trade); always take absolute value.
- `SHROUT` is in thousands.

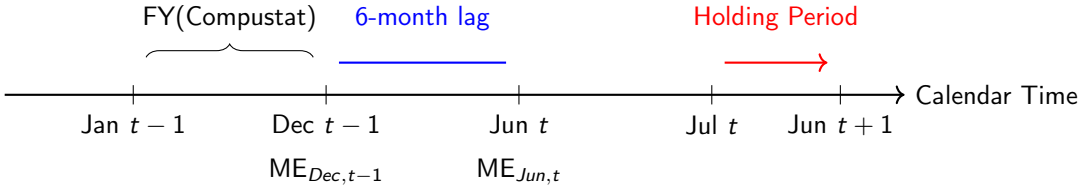
Firm–Level ME (PERMCO Aggregation)

- For firms with multiple share classes (e.g., Alphabet: GOOG and GOOGL), sum `MEQ` across all PERMNOs sharing the same PERMCO and date.
- This gives the total market capitalization of the firm.
- **Critical**: size classification uses firm–level ME, not security–level MEQ.

Delisting Returns

- Delistings are rarely foreseeable; incorporate `DLRET` to avoid downward bias in portfolio returns.
- Adjusted return: $(1 + ret) \times (1 + dlret) - 1$.

Matching CRSP and Compustat: Timeline



Matching Logic

- Compustat `DATADATE` (fiscal year–end) must fall in year $t-1$:

$$\texttt{year(DATADATE)} = t-1$$

- This ensures the accounting data is the most recently available as of June t, with at least a 6–month gap for disclosure.
- CRSP December $t-1$ ME is used for the book–to–market denominator; CRSP June t ME is used for size classification.

NYSE Breakpoints: Why Only NYSE?

Motivation

- NASDAQ joined CRSP in 1972; AMEX in 1962.
- NASDAQ and AMEX stocks tended to be smaller than NYSE stocks.
- Using all exchanges to determine size breakpoints would create a "small" group dominated by NASDAQ firms, breaking consistency over time.

Fama–French Convention

- Size breakpoint: median NYSE market equity in June t.
- B/M breakpoints: 30th and 70th percentiles of NYSE B/M in June t.
- All NYSE, AMEX, and NASDAQ stocks are *assigned* to these NYSE–determined bins.

This treatment became the standard in asset pricing literature.

The 2 × 3 Sort: Portfolio Construction

	Low B/M (Growth)	Medium B/M	High B/M (Value)
Small (S)	S/L	S/M	S/H
Big (B)	B/L	B/M	B/H

Construction Details

- Intersect size (2 groups) and B/M (3 groups) $\Rightarrow$ 6 portfolios.
- Portfolios are **value–weighted** within each cell using lagged market equity as weights.
- All dividends are assumed reinvested proportionally based on market value at the previous month–end.
- Portfolios are rebalanced annually at the end of June.
- Holding period: July of year t through June of year $t + 1$.

From 6 Portfolios to SMB and HML

Definitions

$$SMB_t = \frac{R_{SL} + R_{SM} + R_{SH}}{3} - \frac{R_{BL} + R_{BM} + R_{BH}}{3}$$

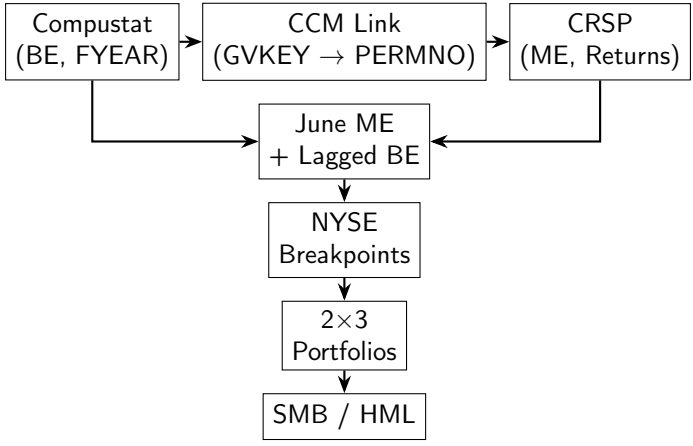
$$HML_t = \frac{R_{SH} + R_{BH}}{2} - \frac{R_{SL} + R_{BL}}{2}$$

Economic Interpretation

- **SMB**: average return of small firms minus average return of big firms, controlling for B/M. Captures the *size premium*.
- **HML**: average return of high B/M (value) firms minus average return of low B/M (growth) firms, controlling for size. Captures the *value premium*.

Note: all portfolio returns are value–weighted; the factors are long–short zero–investment portfolios.

Data Flow: From Raw Data to Factors



- WRDS SAS Code

Common Pitfalls and Implementation Notes

1. **Multi–Class Firms and ME Aggregation**
   - Failing to aggregate MEQ to PERMCO level misclassifies firms like Alphabet as "small" when they are not.
   - Always use firm–level ME for size breakpoints.
2. **Delisting Returns**
   - Omitting `DLRET` creates a downward bias in portfolio returns, especially for small firms with higher delisting rates.
3. **Fiscal Year Alignment**
   - A firm with May fiscal year–end may not have fully disclosed financials by June 30.
   - The 6–month lag is a convention, not a mechanical guarantee.
4. **Primary Share Identification**
   - For firms with multiple share classes, ensure the PERMNO used in CCM corresponds to the primary issue (`LINKPRIM='P'` or `'C'`, or use `sec_history.PRIHISTUSA`).

Summary: Key Takeaways

1. **Never use Ticker or CUSIP** for historical merges; always use permanent identifiers (PERMNO, PERMCO, GVKEY, I/B/E/S TICKER, RavenPack ENTITY_ID).
2. **CCM** (`ccmxpf_lnkhist`) is the gold standard for linking CRSP and Compustat; apply `LINKTYPE`, `LINKPRIM`, and date bounds rigorously.
3. **Primary share** alignment is critical for firms with multiple share classes; use `LINKPRIM` or the new `sec_history` table.
4. **I/B/E/S**: prefer **unadjusted consensus** data; apply CRSP split factors yourself; use summary files for stability across vintages; be explicit about GAAP vs. non–GAAP matching.
5. **RavenPack** adds a real–time news and sentiment layer: use `RELEVANCE >= 90` and `ENTITY_TYPE = 'COMPANY'` for clean firm–level signals; beware source heterogeneity and duplicate stories.
6. **Fama–French replication** requires precise temporal alignment: Dec $t-1$ ME, FY ending in calendar year $t-1$ BE, and June t formation.

Thank you!